

CSE111: ORIENTATION TO COMPUTING

CA2 Report: EDU-Revolution Initiative

1. STUDENT DETAILS

Student Name	ABHINAV KUMAR AMAN
Registration Number	12620906
Roll No.	15
Section	K2P26HR
Term / Semester	First Semester 2026-27
Selected EDU-Revolution Benefit	FutureSkills Prime - Exploratory Data Analysis
Date of Registration / Completion	22 September 2026
Reference ID(s)	Visible on the official portal after registration

2. INTRODUCTION TO EDU-REVOLUTION

EDU-Revolution is a flagship digital skilling initiative integrated into the academic framework to bridge the gap between traditional classroom learning and fast-evolving industry requirements. The primary purpose of this program is to encourage continuous skill acquisition, hands-on technical learning, and self-paced certification through recognized national skilling portals such as FutureSkills Prime (MeitY - Nasscom Digital Skilling Initiative).

Objectives & Significance

- **Skill Alignment:** Aligning fundamental computing and engineering mathematics concepts with high-demand industry skills in Data Analytics, Big Data, and Artificial Intelligence.
- **Practical Learning:** Promoting experiential learning through hands-on dataset manipulation, data cleaning, statistical evaluation, and visual exploration.
- **Employability & Certification:** Enhancing student profiles with government-backed, industry-aligned certification credentials recognized across the IT-ITeS sector.

Integrating EDU-Revolution into the core orientation curriculum ensures that computer science engineering students build real-world analytical capabilities and practical domain expertise alongside their formal university studies.

3. AVAILABLE EDU-REVOLUTION BENEFITS

The EDU-Revolution portal offers several skilling modules mapped to the syllabus, focusing heavily on Data Science, Analytics, and Modern Computing Technologies:

S. No.	Benefit / Initiative	Brief Description
1	Exploratory Data Analysis (Nasscom)	MeitY-Nasscom certification covering quantitative data cleaning, summary statistics, correlation matrices, and visual data exploration.

S. No.	Benefit / Initiative	Brief Description
2	Python for Data Science & Analytics	Hands-on training in Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn for structured data processing.
3	Statistical Methods & Data Mining	Foundational course on probability distributions, hypothesis testing, outlier detection, and data pattern extraction.
4	Data Visualization & Dashboarding	Module focused on building visual insights, graphical plotting, interactive dashboards, and storytelling with data.
5	Big Data Fundamentals & SQL	Introduction to relational databases, SQL queries, data wrangling, and enterprise-scale data processing concepts.
6	Machine Learning Foundations	Introductory training on supervised/unsupervised learning models built upon preprocessed exploratory datasets.

4. SELECTED BENEFIT AND JUSTIFICATION

Selected Benefit Details

- **Course Name:** Exploratory Data Analysis
- **Platform:** FutureSkills Prime (A MeitY - Nasscom Digital Skilling Initiative)
- **Issuing Body:** IT-ITeS Sector Skills Council Nasscom
- **Date of Completion:** 22 September 2026

FutureSkills Prime Certificate of Completion

Figure 1: Official Certificate of Completion for Exploratory Data Analysis issued by FutureSkills Prime (Nasscom / MeitY) to Abhinav Kumar Aman

Description and Key Features of the EDA Course

Exploratory Data Analysis (EDA) is an essential approach in data analysis that employs a variety of quantitative and graphical techniques to maximize insight into a dataset. Key learning outcomes covered in this course include:

- **Data Cleaning & Preprocessing:** Detecting missing values, handling duplicate records, removing noise, and structuring raw unorganized data into tabular formats.
- **Descriptive Statistics:** Calculating measure of central tendency (mean, median, mode), dispersion (variance, standard deviation), and skewness to understand data distribution.
- **Graphical & Visual Analysis:** Utilizing scatter plots, histograms, box plots, heatmaps, and bar charts to identify underlying trends, correlations, and anomalies.
- **Feature Engineering Preparation:** Identifying key variables and dependencies required for downstream machine learning and predictive modeling.

Eligibility and Requirements

This certification is open to undergraduate Computer Science Engineering students seeking formal qualification in data handling, quantitative methods, and analytical programming.

Justification & Relevance to Academic & Career Goals

Data is the backbone of modern computing applications. Selecting Exploratory Data Analysis directly supports my academic progression and future career goals as a CSE core student:

- **Academic Relevance:** Directly complements core coursework including Engineering Mathematics (Linear Algebra, Calculus, Statistics) and Computer Programming (Python, Data Structures) by demonstrating real-world applications of mathematical logic on complex datasets.
- **Career Relevance:** EDA is an mandatory pre-requisite skill for modern tech roles such as Data Engineer, Software Developer, Machine Learning Engineer, and Data Analyst. Mastering EDA equips me with data-driven decision-making capabilities valued by top tech employers.

5. ACTION PLAN

S. No.	Activity / Action	Timeline	Expected Result
1	Portal Registration & Course Enrollment	Week 2	Enrolled in FutureSkills Prime EDA module under EDU-Revolution mapped benefits.
2	Study of Data Cleaning & Preprocessing Modules	Week 3	Learned missing value imputation, outlier detection, and data structuring techniques.
3	Statistical Evaluation & Data Visualization Practice	Week 4	Mastered summary statistics calculation and graphical trend analysis using visual charts.
4	Assessment & Official Certification Clearance	Week 5 (22 Sep 2026)	Successfully cleared the final assessment and received government-aligned Nasscom Certificate.
5	Report Compilation & CA2 Submission	Week 6	Integrated evidence and completed CSE111 CA2 report for final university evaluation.

6. EXPECTED IMPACT

- **Academic Enrichment:** Strengthens mathematical and logical clarity required for analyzing algorithmic performance and engineering data.
- **Skill Development:** Builds hands-on expertise in data cleaning, pattern recognition, graphical visualization, and statistical modeling.
- **Career Preparation:** Provides a MeitY and Nasscom-verified digital credential that adds significant value to technical resumes during internship and campus placement drives.
- **Professional Development:** Develops a structured, data-first mindset essential for solving modern real-world engineering problems.

7. SUPPORTING EVIDENCE & VERIFICATION

Official Certification Details:

- **Candidate Name:** ABHINAV KUMAR AMAN
- **Course Completed:** Exploratory Data Analysis
- **Issuing Authority:** FutureSkills Prime (A MeitY - Nasscom Digital Skilling Initiative)

- **Standard Alignment:** Aligned to Competency Standards developed by IT-ITeS Sector Skills Council Nasscom in collaboration with Industry and Government
- **Date of Issue:** 22 September 2026
- **Verification Status:** Officially verified via QR code & FutureSkills Prime portal registration